

Zhenyu (Tim) YANG

Contact Address

Urban Transport Systems Laboratory
EPFL-ENAC-INTER-LUTS, Station 18
Lausanne, 1015, CH

Personal Information

Email: zhenyu.yang@epfl.ch
Phone: +41 76 752 95 53
 ORCID  Google  GitHub

EDUCATION

National University of Singapore <i>PhD in Industrial and Systems Engineering (ISE)</i> • Thesis: Congestion Management and Information Provision in Risky and Connected Transportation Networks • Advisor: Yang Liu	Singapore 2017-22
Beijing Jiaotong University <i>Master in Urban Traffic Engineering</i>	Beijing 2012-15
Huazhong University of Science and Technology <i>Bachelor in Transportation Engineering</i>	Wuhan 2008-12

EMPLOYMENT HISTORY

Technische Universität München <i>Visiting Researcher (Research group leader, starting from August, 2026)</i> • Advisor: Klaus Bogenberger	Munich 2026-
École Polytechnique Fédérale de Lausanne <i>Lecturer; Postdoctoral Researcher</i> • Advisor: Nikolas Geroliminis	Lausanne 2022-26
National University of Singapore <i>Research Engineer/Fellow</i>	Singapore 2021-22

PUBLICATIONS

Journal Publications

- **Yang, Z.***, Giardina, P., Geroliminis N., & de Palma, A.* (2026). [Recovering scheduling preferences in dynamic departure time models](#), *EPJ Data Science*, 15, 28.
- **Yang, Z.** & Geroliminis, N.* (2025). [Modeling and Regulating Competition between Public Transit and Ride-hailing with Transit Priority Zones](#), *Transportation Research Part C: Emerging Technologies*, 173, 105016
- **Yang, Z.**, de Palma, A., & Geroliminis, N.* (2024). [Tailored Priority Allocation in the Bottleneck Model with General User Heterogeneity](#). *Transportation Research Part B: Methodological*, 190, 103093.
- Stokkink, P.*, **Yang, Z.** (co-first author), & Geroliminis, N. (2025). [Optimal Matching for Ridesharing Systems with Endogenous and Flexible User Participation](#), *Transportation Research Part B: Methodological*, 200, 103290
- Liu, Y.* & **Yang, Z.** (2021). [Information Provision and Congestion Pricing in Risky Road Networks with Heterogeneous Travelers](#). *Transportation Research Part C: Emerging Technologies*, 128, 103083.
- Xie, J., **Yang, Z.**, Lai, X., Liu, Y.*, Yang, X. B., Teng, T. H., & Tham, C. K. (2022). [Deep Reinforcement Learning for Dynamic Incident-Responsive Traffic Information Dissemination](#). *Transportation Research Part E: Logistics and Transportation Review*, 166, 102871.
- Lai, X., **Yang, Z.**, Xie, J., & Liu, Y.* (2024). [Reinforcement Learning in Transportation Research: Frontiers and Future Directions](#). *Multimodal Transportation*, 100164
- Zhou, J., Yi, J.*, **Yang, Z.**, Pu, H., Li, X., Luo, J., & Gao, L. (2024). [A Survey on Vehicle-Drone Collaborative Delivery Operations Optimization: Models, Methods, and Future Research Directions](#). *Swarm and Evolutionary Computation*, 92, 101780
- Zeng, M. H.*, Wang, M., Chen, Y., & **Yang, Z.** (2021). [Dynamic Evacuation Optimization Model Based on Conflict-Eliminating Cell Transmission and Split Delivery Vehicle Routing](#). *Safety Science*, 137, 105166.

Under Review

- **Yang, Z.**, Liu, Y.*, & Geroliminis, N. (2026). [Optimal Routing of Mixed Autonomy Traffic Flow with Uniform Tolling under Attentional Costs](#), submitted to *Transportation Science*

- Jin, R., Yang, Z.Q., Ji, R., **Yang, Z.**, Sun, K., & Dong, H.* (2025). A Spatiotemporal Diffusion Model with Graph and Dilated Convolution for Faster Traffic Data Imputation, submitted to *Expert Systems With Applications*
- Yang, Z.Q., Jin, R., **Yang, Z.**, & Dong, H.* (2025) A Dual-Branch Multisensor Fusion Framework with Multicomponent Channel Attention for High-Speed Train Traction System State Prediction, submitted to *IEEE Transactions on Industrial Informatics*

Working Papers Close to Completion

- **Yang, Z.** & Liu, Y.* Optimal Traffic Information Provision with Rational Inattentive Travelers, to be submitted to *Production and Operations Management*
- Wang, W., **Yang, Z.***, Gao, Z., & Geroliminis, N. On the Modeling and Solution of Drone-Truck Cooperative Delivery with Restrictive Areas, to be submitted to *Transportation Research Part B: Methodological*

Selected Work in Progress

- Zhu, D., **Yang, Z.***, & Geroliminis, N. On the Decentralized Ridesharing Organization with Autonomous Vehicles and Priority Schemes, work in progress
- Li, Y., & **Yang, Z.***, & Geroliminis, N. Optimal Location-Based Parking Pricing Design for Private Parking Sharing Problem Considering Stable Matching, work in progress
- de Palma, A., Xu, Meng, Geroliminis, N., & **Yang, Z.***, Vulnerability Assessment of Transport Networks under Extreme Conditions, work in progress

RESEARCH PROJECT EXPERIENCE

MOTVS: Human-Oriented Ubiquitous Mobility	2024-present
• Funding Source: Swiss National Science Foundation • Led the proposal development; secured a grant of 478,720 CHF • Developed SP and RP surveys and designed a novel preference estimation method based on RP data	
Opt-USE: Optimizing the Physical Workplace for Urban Socio-Environmental Performance	2024-present
• Funding Source: ENAC Fribourg Grant • Co-led proposal development; secured a grant of 250,000 CHF • Modelling the impact of flexible working policies on social, spatial, energy, and transportation systems	
DIT4TraM: Distributed Intelligence & Technology for Traffic & Mobility Management	2022-24
• Funding Source: EU Horizon 2020 • Developed metering-based priority schemes for congestion management at road bottlenecks • Prepared the deliverable reports (as the first-author) for two WPs in collaboration with project partners	
Urban Traffic Flow Smoothening	2021-22
• Funding Source: STE-NUS Joint Lab • Integrated reinforcement learning-based traffic control methods with a simulation environment based on Vissim • Collaborated with the industrial partner STE and delivered the developed codebase	

PRESENTATIONS AND INVITED TALKS

Conference Presentations

- **Yang, Z.**, Giardina, P., de Palma, A., & Geroliminis, N.* (2026). Reverse ADL-Vickrey: Inferring Scheduling Preferences based on Congestion and Choice Data, In *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., USA
- **Yang, Z.**, Giardina, P., de Palma, A., & Geroliminis N.* (2025). Reverse ADL-Vickrey: Inferring Scheduling Preferences based on Congestion and Choice Data, In *25th Swiss Transport Research Conference*, Ascona, Switzerland
- **Yang, Z.**, de Palma, A., & Geroliminis N.* (2025). Modeling and Regulating Competition between Public Transit and Ride-hailing with Geofencing Policy, In *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., USA
- Stokkink, P.*, **Yang, Z.**, & Geroliminis, N. (2025). Optimal Matching and Subsidy Strategies for Ridesharing Systems with Endogenous and Flexible User Participation, In *12th Symposium of the European Association for Research in Transportation*, Helsinki, Finland
- Dong, H.*, Liu, K., **Yang, Z.**, Li, Z., Jia, L., Qin, Y., & Wang, Y. (2017). Understanding Spatial and Temporal Patterns of Urban Travel Demands with Call Detail Record Data. In the *96th Annual Meeting of the Transportation Research Board*, Washington, D.C., USA

- Zhang, C., Dong, H.*, Jia, L., Qin, Y., & **Yang, Z.** (2017). Robust Vehicle Detection and Identification with Single Magnetic Sensor. In *2017 2nd IEEE International Conference on Intelligent Transportation Engineering*. Singapore
- **Yang, Z.**, & Ge, Z.* (2014) An Evolution Model of the Express Freight Network. In *2nd International Conference on Civil Engineering, Architecture and Sustainable Infrastructure*. Zhengzhou, China
- Lin, M.*, Li, Y. & **Yang, Z.** (2014). “转型期汕头市城市交通战略探索”[A Study on the Transport Development Strategy of Shantou in the Transitioning Period]. In *新型城镇化与交通发展——2013 年中国城市交通规划年会 (the 2013 China Urban Transportation Planning Annual Conference)*. Fuzhou, China

Invited Talks

- Understanding and Managing Commuter Heterogeneity: From Preference Inference with RP Data to Heuristic Priority Schemes, T&L colloquium, TU Delft, Delft, the Netherlands, 2026
- Reverse ADL-Vickrey: Inferring Scheduling Preferences based on Congestion and Choice Data, AFFINITE Seminar, THEMA, Paris, France, 2025
- Modeling and Application of Road Space Priority Allocation Strategies in Transportation Demand Management, Tongji University, Shanghai, China, 2024
- Tailored Priority Allocation in the Bottleneck Model with General User Heterogeneity, 25th COTA Winter Symposium, Washington D.C., USA, 2023

TEACHING

Lecturer at EPFL (Primary course instructor)

CIVIL455 Transportation Economics

Spring 2023-26

Guest Lecturer at EPFL

CIVIL457 Fundamentals of Traffic Operations and Control

Spring 2024

Teaching Assistant at NUS

CE5209 Data Analytics and Modeling

Fall 2021

IE5121 Quality Planning and Management

Spring 2018-20

IE5001 Operations Planning and Control I

Fall 2018-19

IE4221 Transportation Demand Modeling and Economics

Fall 2017

INTERNSHIPS

Urban Planning & Design Institute of Shenzhen (UPDIS)

Transportation Planning Intern

Shenzhen
Summer 2011-13

National Development and Reform Commission

Transportation Planning Intern, Institute of Comprehensive Transportation

Beijing
2013-14

China Academy of Urban Planning and Design

Editorial Intern, Urban Transport of China

Beijing, China
2013

HONORS

Selected for an untenured Associate Professor position at Beijing Jiaotong University

2026

Finalist, 2021 INFORMS Annual Meeting Poster Competition

2021

NUS Research Scholarship

2017

Huiguang Cup of Paper Competition at Beijing Jiaotong University

2014

Outstanding Graduate, Huazhong University of Science and Technology

2012

Hilti Scholarship

2010

Excellent Student Cadre Scholarship

2009

Excellent Freshman Scholarship

2009

PROFESSIONAL SERVICES

Reviewer for Transportation Research Part B, C, E, Transport Policy, Transportmetrica A, and Physica A

Head, Local Organising Committee, Conference on Emerging Technologies in Transportation Systems (TRC-30)

Member, Local Organising Committee, 26th International Symposium on Transportation and Traffic Theory (ISTTT 26)

Member, Local Organising Committee, 3rd International Symposium on Multimodal Transportation (ISMT 2019)

Young Member, Technical Committee, World Transport Congress (WTC)

Member, Chinese Overseas Transportation Association (COTA)

Last updated: July 18, 2026

REFERENCES

Nikolas Geroliminis

Professor

Urban Transport Systems Laboratory

École Polytechnique Fédérale de Lausanne

Email: nikolas.geroliminis@epfl.ch

Website: <https://people.epfl.ch/nikolas.geroliminis>

Yang Liu

Associate Professor

Department of Civil and Environmental Engineering & Department of Industrial Systems Engineering and Management

National University of Singapore

Email: iseliuy@nus.edu.sg

Website: <https://cde.nus.edu.sg/sem/staff/liu-yang/>

André de Palma

Senior Excellence Chair in Economics

CY Cergy Paris Université, THEMA

Email: andre.de-palma@cyu.fr

Website: <https://thema.u-cergy.fr/membres/enseignants-chercheurs/de-palma?lang=fr>

Michel Bierlaire

Professor

Transportation and Mobility Laboratory

École Polytechnique Fédérale de Lausanne

Email: michel.bierlaire@epfl.ch

Website: <https://people.epfl.ch/michel.bierlaire/?lang=en>

Qiang Meng

Professor

Department of Civil and Environmental Engineering

National University of Singapore

Email: ceemq@nus.edu.sg

Website: <https://cde.nus.edu.sg/cee/staff/meng-qiang/>